Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Examination, 2020

URP 2291

(Statistics for Planners II)

Full Marks: 120 Time: 1.5 hrs

N.B. i) Answer any **two** questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section A

1. (a) Explain the concept of normal distribution. Justify with appropriate example that normal distribution is a natural tendency of large dataset. (09)

(b) Suppose the mean CGPA of URP-18 is 3.33 with a standard deviation of 0.3. If the CGPA is normally distributed among 60 students and your CGPA is 3.63, interpret your class position. (12)

(c) What is the basic difference between z and t distribution? Explain graphically. (09)

2. (a) A pharmaceutical company is measuring the effectiveness of a new drug to see whether the drug is effective in reducing high blood pressure. Write the null hypothesis and alternate hypothesis. (06)

(b) Write short notes on the followings: (24)

(i) One-tailed test vs. two-tailed test,

(ii) Level of significance,

(iii) Critical region,

(iv) Statistical significance vs. clinical significance.

3. (a) Prove that chi-square follows a binomial distribution. (12)

(b) An investor is planning to buy a restaurant. The owner of the restaurant claims that on an average there are 500 customers per week (six days a week). Day-wise proportion (%) is shown in column (b) of below table. However, the potential investor makes a survey of a week and finds the number of customers as shown in column (c). (18)

Is the claim of owner of the restaurant genuine/true?

Day (a)	Owners claim of % of customer (b)	Investors survey of the number of customer (c)
Wednesday	15	60
Thursday	10	35
Friday	35	120
Saturday	30	110
Sunday	5	40
Monday	5	50

Section B

4. (a) Illustrate the concept of simple, multiple and partial correlation with examples. (08)
   (b) Why do you think correlation fails to draw causation? Explain in brief. (07)
   (c) From the following table establish a linear regression model of number of houses on number of population. (15)

No. of houses	2	4	5	3	6	8	7
No. of population	20	28	30	15	40	32	29

5. (a) Without shifting of origin, show that regression co-efficient depends on scale of measurement. (10)
   (b) The following results of the rainfall (x) and crop production (y) of 100 different years of a region are given: (13)

   Mean (x) = 50 cm
   Mean (y) = 120 tons
   Standard Deviation (x) = 3 cm
   Standard Deviation (y) = 15 tons

   The coefficient of correlation between x and y is 0.8. Estimate the most likely crop production for a rainfall of 100 cm.

   (c) Discuss the concept of least square method with necessary example. (07)

6. (a) Why is it necessary to smooth out time series data? Give your logical reasoning in short. (07)
   (b) Smooth out the following time series data of quarterly car sales using a 4 year moving average. Deseasonalize the car sales data and forecast the sales for year 3. (23)

Year	Quarter	Sales (1000$)
1	1	4.8
	2	4.1
	3	6.0
	4	6.5
Year		
2	1	5.8
	2	5.2
	3	6.8
	4	7.4

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Examination, 2020

URP 2251

(Landscape Planning and Design)

Full Marks: 120 | Time: 1.5 hrs

N.B. i) Answer any **two** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section A

1. (a) Explain the importance of studying landscape planning and design in Urban and Regional Planning. (15)
   (b) Draw different types of line indicating their use in the sketch of a sample landscape layout plan. (15)

2. (a) Explain different types of forms in landscape planning and design. (15)
   (b) Briefly describe the Carbon cycle with necessary drawing. (15)

3. (a) Proportion is an important principle of landscape planning and design. Evaluate this statement with a necessary drawing. (15)
   (b) Explain different uses of plants in landscape planning and design (15)

Section B

4. (a) Define site and context. How can site analysis ensure a contextual landscape planning and design? Analyze with example. (15)
   (b) Write short notes (any 3): (15)
      i) View and Vista
      ii) Symmetricity in landscape
      iii) The Axis
      iv) The Terminus

5. (a) Define landscape ecology and ecosystem. Discuss the historical perspective of landscape ecology with examples. (15)
   (b) Illustrate the steps in residential landscape design process with relevant examples. (15)

6. (a) Draw an explanatory connection among residential, commercial and recreational landscaping with necessary sketches. (15)
   (b) Define landscape conservation. Briefly describe the principles of conservation landscape. (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2020
Hum 2217
(Public and Local Government Finance)

Full Marks: 120 Time: 1.5 hrs

N.B. i) Answer any **TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define public finance. Distinguish between public finance and private finance. (10)
   (b) Describe the importance of studying public finance in urban and regional planning discipline. (10)
   (c) Discuss the main functions of public finance of a country. (10)

2. (a) What are the main resources of revenue of Bangladesh Government? Discuss. (15)
   (b) Discuss the canons of taxation. (15)

3. (a) Explain the following terms: (15)
   (i) Tax impact (ii) Public debt (iii) Free rider problem (iv) Value added tax
   (b) Write about the methods of redemption of public debt. (15)

Section – B

4. (a) What is budget? (05)
   (b) Discuss the objective of a good budget. (10)
   (c) Describe the steps of budgetary process. (15)

5. (a) What are the basic differences between deficit financing and deficit budgeting? (08)
   (b) How can deficit financing be used during the depression of an economy? (10)
   (c) Discuss the steps of formulating government budget. (12)

6. (a) What do you mean by local government? Explain the structure of local government of Bangladesh. (08)
   (b) Describe the formation system of zilla parishad. (10)
   (c) State the structure and functions of city corporation as a local government of Bangladesh. (12)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Regular Examination, 2020

URP2281

(GIS and Remote Sensing)

Full Marks: 120 Time: 1.5 hrs

N.B. i) Answer any **TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) "Geospatial data consists of information systems and geographical positions" – Explain the statement with an example related to spatial planning problems in your community. (12)

   (b) GIS can be applied to combat COVID-19 transmission. Discuss the applicability of GIS to different aspects (i.e., location, condition, trend, routing, pattern, and modeling) to mitigate the COVID-19 challenges in Bangladesh. (18)

2. (a) Why are spatial references (i.e., coordinate systems and map projection) in data sets important for consistent analysis? Explain with rationale examples. (15)

   (b) Suppose you have paper mouza maps of Khulna for some housing projects. You need to link your proposed plans to the mouza map for implementing the projects. Is it necessary to georeference the paper maps before performing any geoprocessing to link the proposed plans? If so, describe the process. If not, explain why? (15)

3. (a) Suppose you have converted the geographic data of your community, either from a hardcopy or a scanned image, into vector data by tracing the features. What kind of errors could occur during processing? Discuss in brief with necessary examples and figures. (12)

   (b) Assume you have physical data (vector data) for your community that you want to use to solve some spatial planning problems. Describe the applicability of different spatial operations for solving the problems in your community. Use practical examples and figures where necessary. (18)

Section – B

4. (a) Briefly explain the importance of Remote Sensing in the field of Urban and Regional Planning. (18)

   (b) Differentiate among Ground-based, Airplane-based, and Satellite-based remote sensing platforms based on their respective applications. (12)

5. (a) "High spatial resolution satellite image is not the only true indicator to reflect high quality, rather it depends on other types of resolutions especially radiometric resolution" – explain with a necessary example. (15)

   (b) Compare two sets of satellite Images having different types of different resolutions in the following based on their applications. Explain with the necessary example. (15)

Satellite Image (A)	Satellite Image (B)
• High spatial resolution (0.3 - 4 m) • Low spectral resolution (1- 3 bands) • High radiometric resolution (16 bit and above)	• High spatial resolution (0.3 - 4 m) • High spectral resolution (16- 220 bands) • High radiometric resolution (16 bit and above)

6. (a) Suppose, you are a GIS and Remote Sensing professional, and going to purchase high-resolution satellite imagery for a development plan preparation project for extracting features as an alternative to the physical survey to save time and resources. What are the issues/specifications you will check from the technical point of view to fulfill the project's need for the satellite imageries before the decision to purchase and why? (18)

   (b) Assume that the distance between two points was measured to be 120 mm on a vertical photograph and 300 mm on a map. If the surveying ground distance between the same two points is 6000 m, what will be the photo scale and map scale and the scale reciprocals of the photograph and the map? Provide your judgments on the photo scale and map scale to describe the level of detailing of the photographs and their respective application. (12)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Regular Examination, 2020

CE 2241

(Principles of Environmental Engineering)

Full Marks: 120 Time: 1.5 hrs

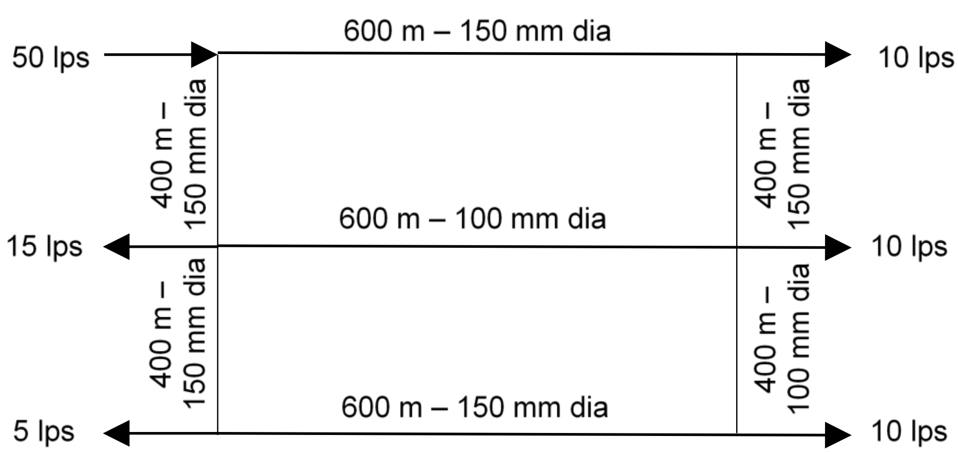
N.B. i) Answer any TWO questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Show the efficiency of a settling tank is dependent on surface area, not depth of tank. (08)
(b) Write down the main sources of fresh water. In which source do you prefer for Khulna city – Describe briefly. (10)
(b) A 100 mm diameter tube well produces 70 l/s with a drawdown of 4 m and the circle of influence of 100 m in diameter. The static depth of water in the well is 42 m. Calculate the coefficient of permeability of the aquifer in which the tube well is sunk. (12)

2. (a) What is water demand? Write down the factors that affect the rate of water demand. (08)
(b) Write down the methods of water distribution system. In which distribution system is suitable for KUET campus – Explain. (10)
(c) The population of a city is 5,000 and the city covers 400 m^2 public park and 1000 m^2 street. Determine the water demand of the city. (12)

3. (a) Explain the coagulation-flocculation process. How does it differ from the plain sedimentation? (10)
(b) Calculate the flow in each of the pipes in the following looped pipe network (at least two trial). (20)



Section – B

4. (a) What is sanitation system? Explain the role of sanitation in controlling the transmission of excreta-related diseases. (10)
(b) Draw a schematic diagram showing various components of a septic tank. How the position of inlet and outlet devices can influence the septic tank operation. (10)
(c) Design leach pits for a twin off-set pour-flush sanitation unit for a family of 5 members having a life period of 3 years. The average wastewater flow rate is about 9 liters/person/day. The soil is porus silty loam with a long term infiltration rate of 30 l/m^2 day. (10)

5. (a) How to reduce the air pollution in Khulna city – explain the techniques. (10)
(b) Write down the solid waste collection methods. In which method do you think suitable for Khulna city? (10)
(c) Briefly explain the symbiosis between algae and bacteria. Describe the bacterial growth with sketch. (10)

6. (a) Distinguish between storm drains and sewers. Why is drainage system for a roadway traversing an urbanized region more complex the roadway traversing sparsely settled rural areas? (10)

(b) Which method is more preferable to measure the momentary peak flow rate of storm drains and why? How can secondary pollutant can be formed? (10)

(c) Calculate the cannel travel time (t_t), contributing area (A_c) and gutter flow (Q) by Manning's equation for the design of storm drainage by using following data: (10)
L = length of pipe, 100 ft, V = velocity 25 ft/sec, A = ft^2, area of basin = 15 ft^2, t_{c1} = 40 min, t_{c2} = 60 min. S_x = cross slope 0.78 ft/ft, S = longitudinal slope 1.20 ft/ft, T= spread 150 ft, n= 0.150, Manning's n.
